

UNIVERSIDAD DE LEÓN

Grado en Ingeniería de Datos e IA

Escuela de Ingenierías Industrial, Informática y Aeroespacial



CRYPTOLEARNING COURSE

Aplicación para la enseñanza y aprendizaje de Activos digitales

Entrega 3: Modelo de Diseño

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1. Diagramas de Interacción (Colaboración o secuencia).

1.1. CU1: Iniciar Sesión.

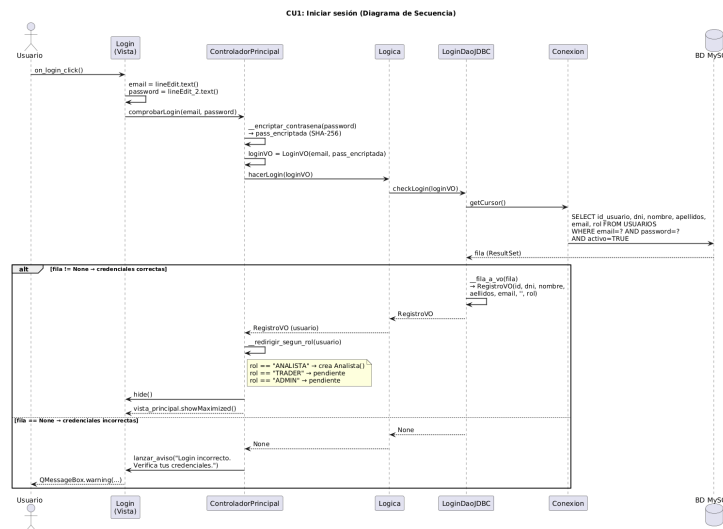


Figura 1: CU1

1.2. CU2: Iniciar Sesión y CU3: Registrarse y asignar saldo inicial.

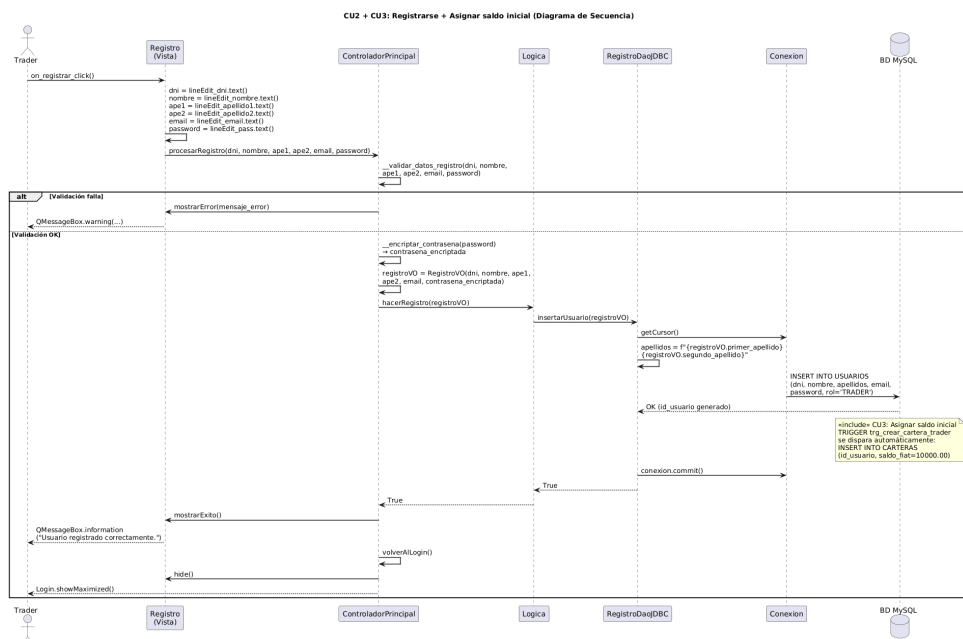


Figura 2: CU2 y CU3

1.3. CU4: Consultar criptomonedas.

CU4: Consultar criptomonedas (Diagrama de Colaboración)

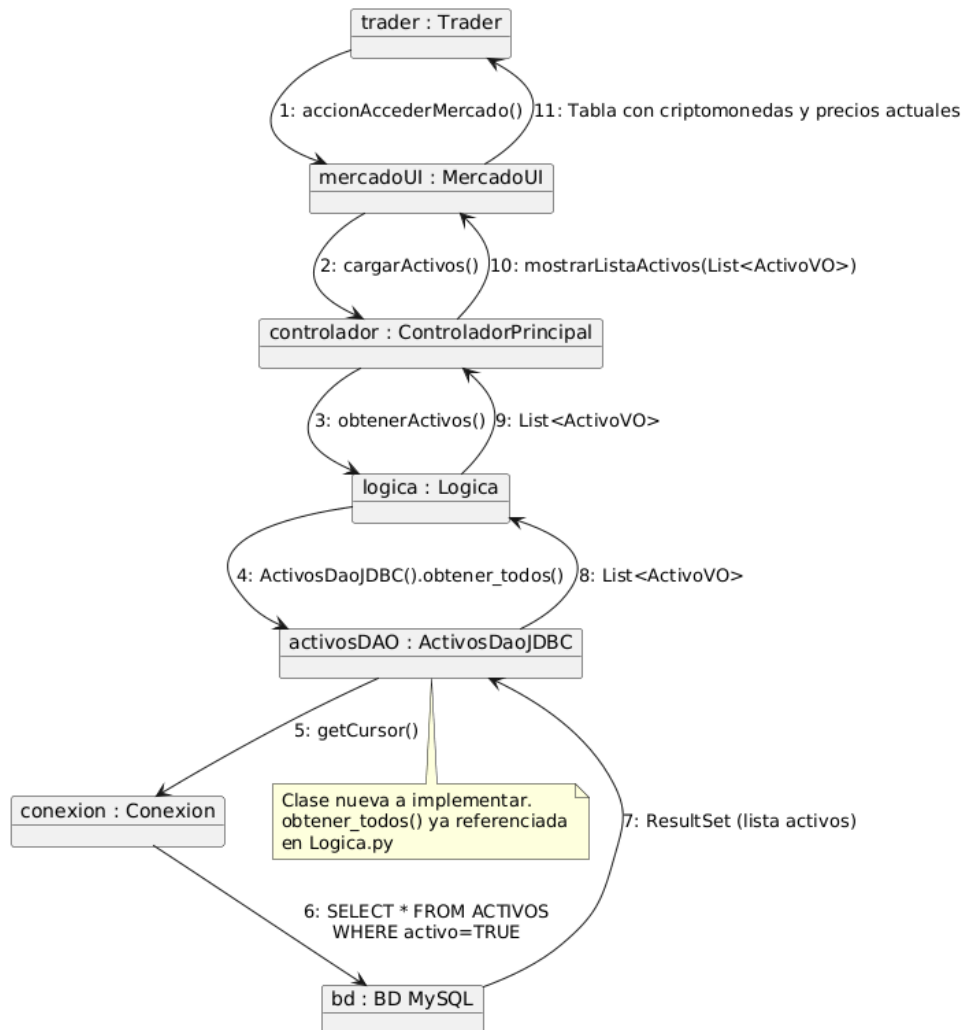


Figura 3: CU4

1.4. CU5: Ver detalles y gráfico.

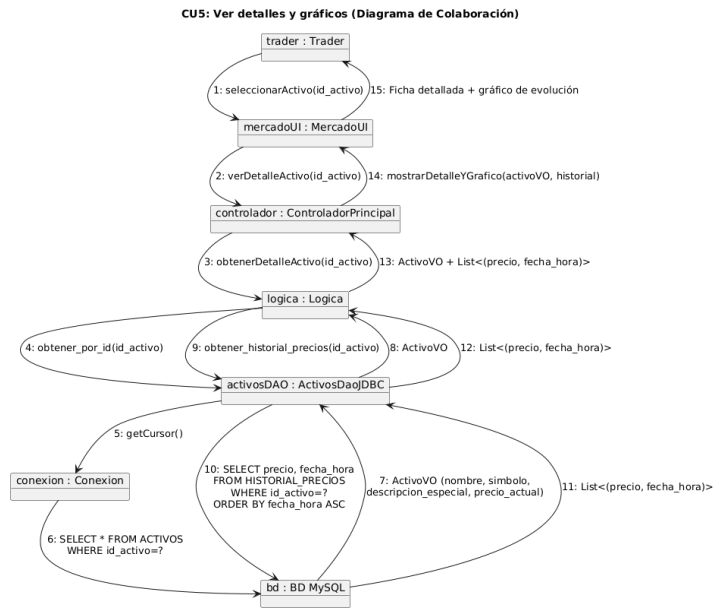


Figura 4: CU5

1.5. CU6: Comprar/vender activos, CU7: Verificar saldo y CU8: Registrar operación

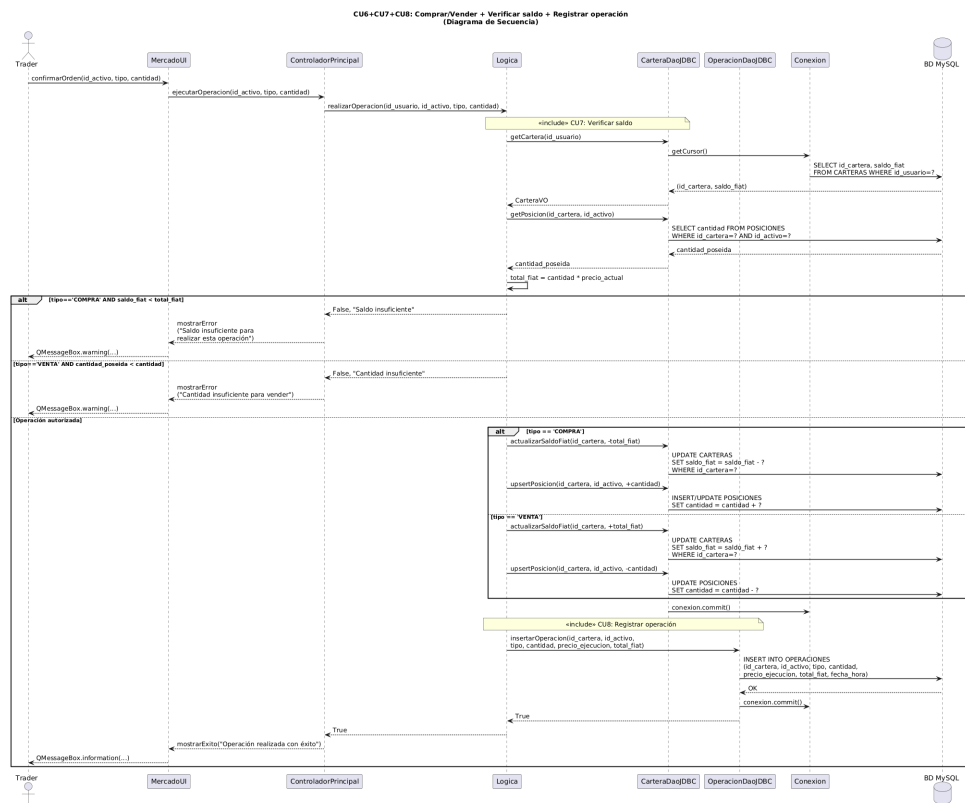


Figura 5: CU6, CU7 y CU8

1.6. CU9: consultar cartera y saldo

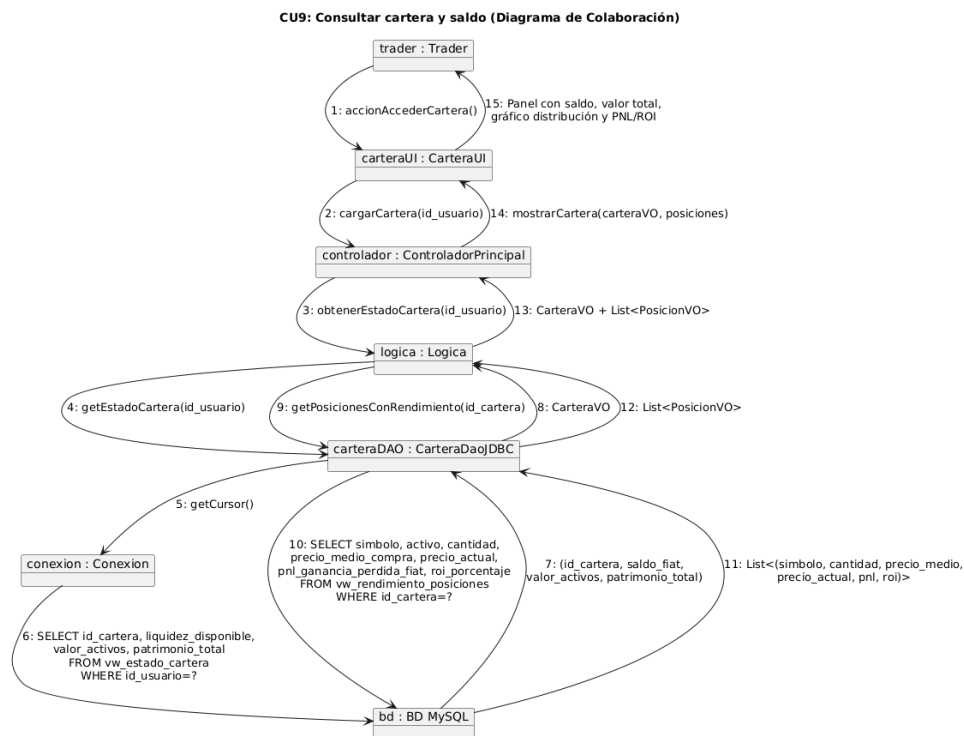


Figura 6: CU9

1.7. CU10: Leer noticias e informes.

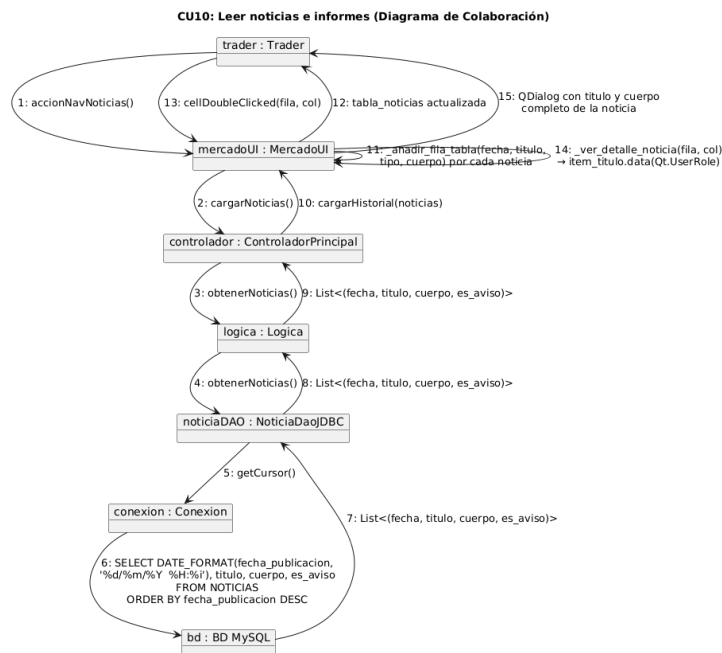


Figura 7: CU10

1.8. CU11: Publicar noticias e informes.

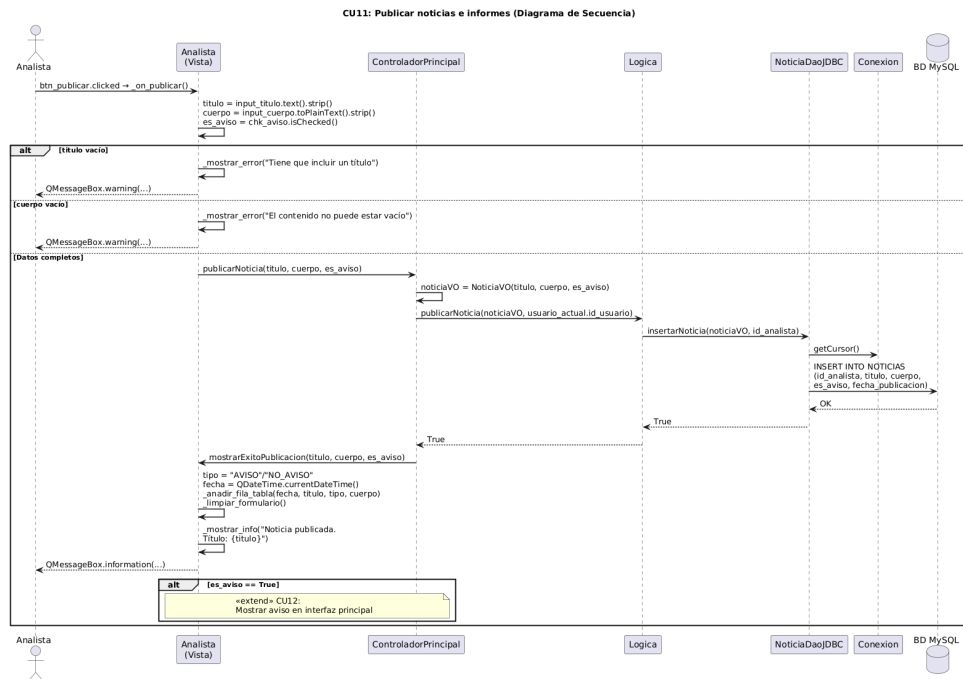


Figura 8: CU11

1.9. CU12: Mostrar aviso en interfaz principal.

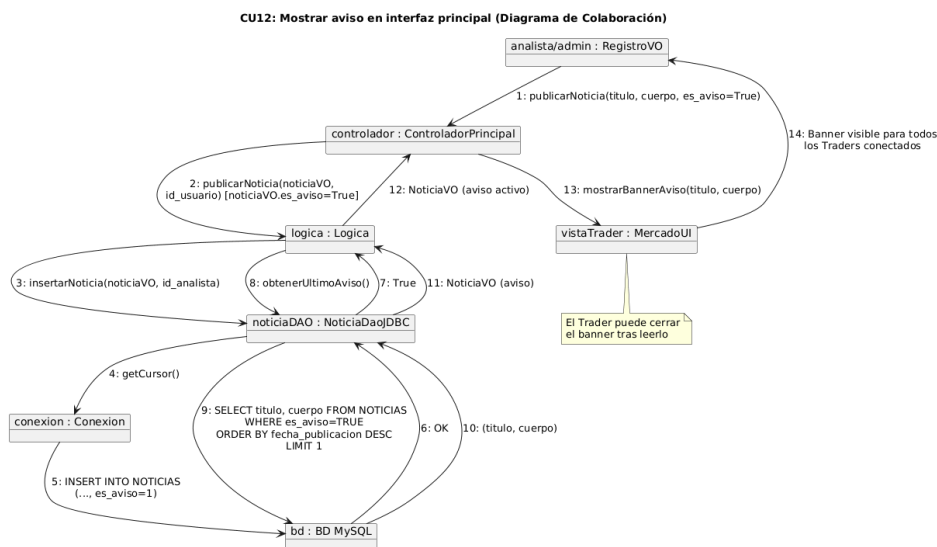


Figura 9: CU10

1.10. CU13: Gestionar usuarios y CU14: Modificar o eliminar usuarios.

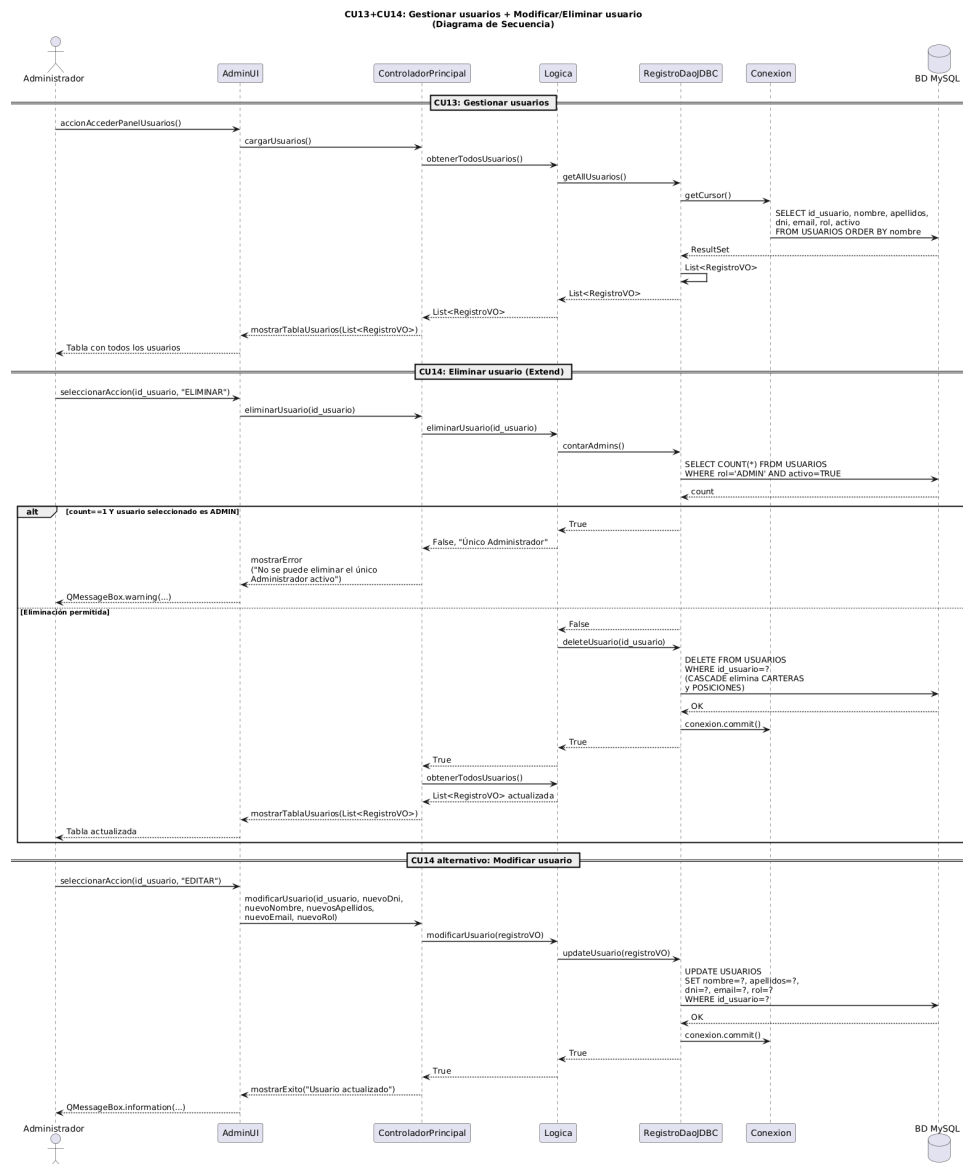


Figura 10: CU13 y CU14

1.11. CU15: Retirar/Eliminar criptomonedas.

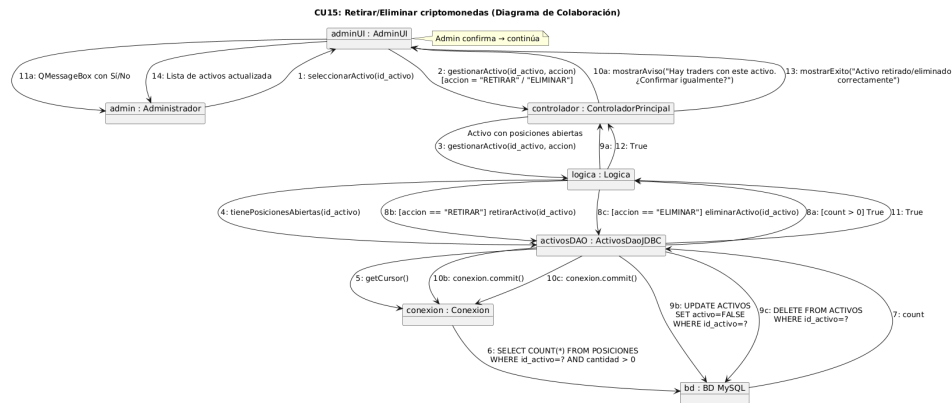


Figura 11: CU15

1.12. CU17: Publicar evento de mercado.

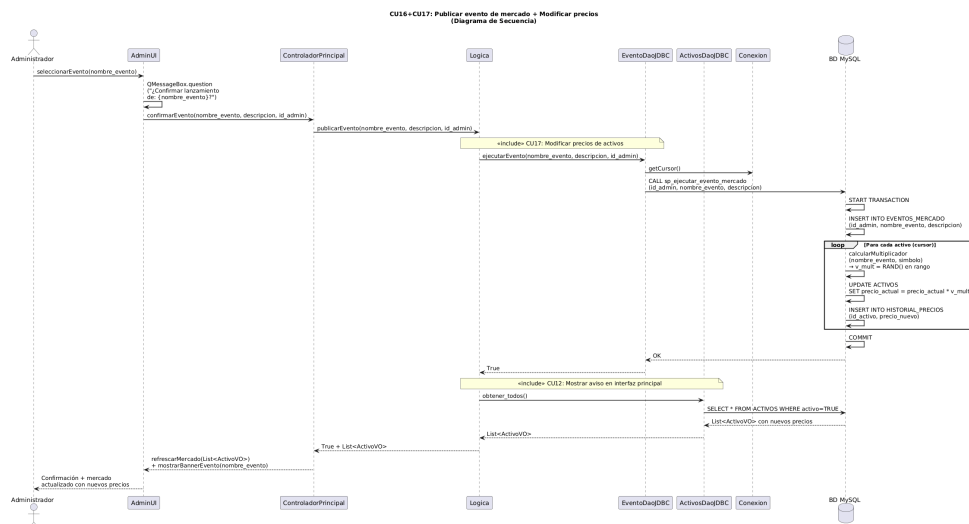


Figura 12: CU16 y CU17

The diagram illustrates the architecture of a Human Resource Management System, organized into three main layers: **Entidad** (Entities), **Controlador** (Controllers), and **Interfaz** (Interfaces).

Entidad (Entities): These classes represent the core data objects of the system.

- UsuarioVO**: Attributes include `_id` (int), `_dni` (string), `_password` (string), `_email` (string), `_telefono` (string), and `_direccion` (string).
- EmpleadoVO**: Attributes include `_id` (int), `_dni` (string), `_password` (string), `_email` (string), `_telefono` (string), `_direccion` (string), `_fecha_nacimiento` (date), `_fecha_contratacion` (date), `_fecha_baja` (date), `_estado` (boolean), `_cargo` (string), `_salario` (float), `_horario` (string), `_observaciones` (string), and `_fecha_ultima_actualizacion` (date).
- AdministradorVO**: Attributes include `_id` (int), `_dni` (string), `_password` (string), `_email` (string), `_telefono` (string), `_direccion` (string), `_fecha_nacimiento` (date), `_fecha_contratacion` (date), `_fecha_baja` (date), `_estado` (boolean), `_cargo` (string), `_salario` (float), `_horario` (string), `_observaciones` (string), and `_fecha_ultima_actualizacion` (date).
- TrainerVO**: Attributes include `_id` (int), `_dni` (string), `_password` (string), `_email` (string), `_telefono` (string), `_direccion` (string), `_fecha_nacimiento` (date), `_fecha_contratacion` (date), `_fecha_baja` (date), `_estado` (boolean), `_cargo` (string), `_salario` (float), `_horario` (string), `_observaciones` (string), and `_fecha_ultima_actualizacion` (date).
- RegistroVO**: Attributes include `_id` (int), `_dni` (string), `_password` (string), `_email` (string), `_telefono` (string), `_direccion` (string), `_fecha_nacimiento` (date), `_fecha_contratacion` (date), `_fecha_baja` (date), `_estado` (boolean), `_cargo` (string), `_salario` (float), `_horario` (string), `_observaciones` (string), and `_fecha_ultima_actualizacion` (date).
- LogVO**: Attributes include `_id` (int), `_dni` (string), `_password` (string), `_email` (string), `_telefono` (string), `_direccion` (string), `_fecha_nacimiento` (date), `_fecha_contratacion` (date), `_fecha_baja` (date), `_estado` (boolean), `_cargo` (string), `_salario` (float), `_horario` (string), `_observaciones` (string), and `_fecha_ultima_actualizacion` (date).
- NoticiaVO**: Attributes include `_id` (int), `_dni` (string), `_password` (string), `_email` (string), `_telefono` (string), `_direccion` (string), `_fecha_nacimiento` (date), `_fecha_contratacion` (date), `_fecha_baja` (date), `_estado` (boolean), `_cargo` (string), `_salario` (float), `_horario` (string), `_observaciones` (string), and `_fecha_ultima_actualizacion` (date).
- OperadorVO**: Attributes include `_id` (int), `_dni` (string), `_password` (string), `_email` (string), `_telefono` (string), `_direccion` (string), `_fecha_nacimiento` (date), `_fecha_contratacion` (date), `_fecha_baja` (date), `_estado` (boolean), `_cargo` (string), `_salario` (float), `_horario` (string), `_observaciones` (string), and `_fecha_ultima_actualizacion` (date).
- EventoVO**: Attributes include `_id` (int), `_dni` (string), `_password` (string), `_email` (string), `_telefono` (string), `_direccion` (string), `_fecha_nacimiento` (date), `_fecha_contratacion` (date), `_fecha_baja` (date), `_estado` (boolean), `_cargo` (string), `_salario` (float), `_horario` (string), `_observaciones` (string), and `_fecha_ultima_actualizacion` (date).
- PartidoVO**: Attributes include `_id` (int), `_dni` (string), `_password` (string), `_email` (string), `_telefono` (string), `_direccion` (string), `_fecha_nacimiento` (date), `_fecha_contratacion` (date), `_fecha_baja` (date), `_estado` (boolean), `_cargo` (string), `_salario` (float), `_horario` (string), `_observaciones` (string), and `_fecha_ultima_actualizacion` (date).
- OperacionVO**: Attributes include `_id` (int), `_dni` (string), `_password` (string), `_email` (string), `_telefono` (string), `_direccion` (string), `_fecha_nacimiento` (date), `_fecha_contratacion` (date), `_fecha_baja` (date), `_estado` (boolean), `_cargo` (string), `_salario` (float), `_horario` (string), `_observaciones` (string), and `_fecha_ultima_actualizacion` (date).

Controlador (Controllers): These classes manage the logic and data flow between the entities and the interfaces.

- ControladorPrincipal**: The main controller, responsible for managing the system's state and coordinating the other controllers. It has a `ControladorPrincipal` interface and a `ControladorPrincipal` class.
- ControladorUsuario**: Manages user-related operations. It has a `ControladorUsuario` interface and a `ControladorUsuario` class.
- ControladorEmpleado**: Manages employee-related operations. It has a `ControladorEmpleado` interface and a `ControladorEmpleado` class.
- ControladorTrainer**: Manages trainer-related operations. It has a `ControladorTrainer` interface and a `ControladorTrainer` class.
- ControladorRegistro**: Manages registration-related operations. It has a `ControladorRegistro` interface and a `ControladorRegistro` class.
- ControladorLog**: Manages login-related operations. It has a `ControladorLog` interface and a `ControladorLog` class.
- ControladorNoticia**: Manages news-related operations. It has a `ControladorNoticia` interface and a `ControladorNoticia` class.
- ControladorOperador**: Manages operator-related operations. It has a `ControladorOperador` interface and a `ControladorOperador` class.
- ControladorEvento**: Manages event-related operations. It has a `ControladorEvento` interface and a `ControladorEvento` class.
- ControladorPartido**: Manages party-related operations. It has a `ControladorPartido` interface and a `ControladorPartido` class.
- ControladorOperacion**: Manages operation-related operations. It has a `ControladorOperacion` interface and a `ControladorOperacion` class.

Interfaz (Interfaces): These classes define the user interface and the data exchange between the controllers and the entities.

- InterfazUsuario**: Defines the user interface for user management. It has a `InterfazUsuario` interface and a `InterfazUsuario` class.
- InterfazEmpleado**: Defines the user interface for employee management. It has a `InterfazEmpleado` interface and a `InterfazEmpleado` class.
- InterfazTrainer**: Defines the user interface for trainer management. It has a `InterfazTrainer` interface and a `InterfazTrainer` class.
- InterfazRegistro**: Defines the user interface for registration management. It has a `InterfazRegistro` interface and a `InterfazRegistro` class.
- InterfazLog**: Defines the user interface for login management. It has a `InterfazLog` interface and a `InterfazLog` class.
- InterfazNoticia**: Defines the user interface for news management. It has a `InterfazNoticia` interface and a `InterfazNoticia` class.
- InterfazOperador**: Defines the user interface for operator management. It has a `InterfazOperador` interface and a `InterfazOperador` class.
- InterfazEvento**: Defines the user interface for event management. It has a `InterfazEvento` interface and a `InterfazEvento` class.
- InterfazPartido**: Defines the user interface for party management. It has a `InterfazPartido` interface and a `InterfazPartido` class.
- InterfazOperacion**: Defines the user interface for operation management. It has a `InterfazOperacion` interface and a `InterfazOperacion` class.

The diagram shows a complex network of relationships between these classes, including inheritance, association, and aggregation. The **ControladorPrincipal** class is the central hub, coordinating the system's operations. The **Interfaz** classes provide the user interface for the system, while the **Controlador** classes manage the logic and data flow.

Figura 13: Diagrama de clases de diseño

3. Diagrama de paquetes

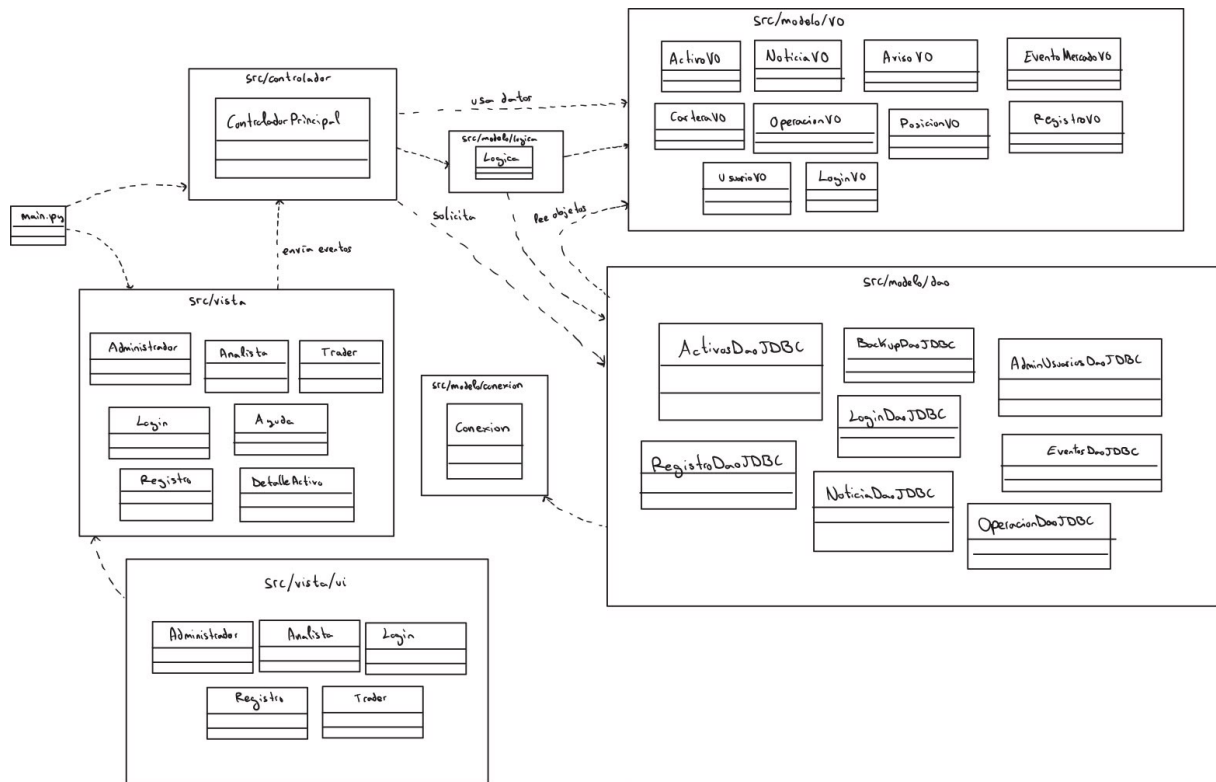


Figura 14: Diagrama de paquetes

4. Esquema relacional de la base de datos

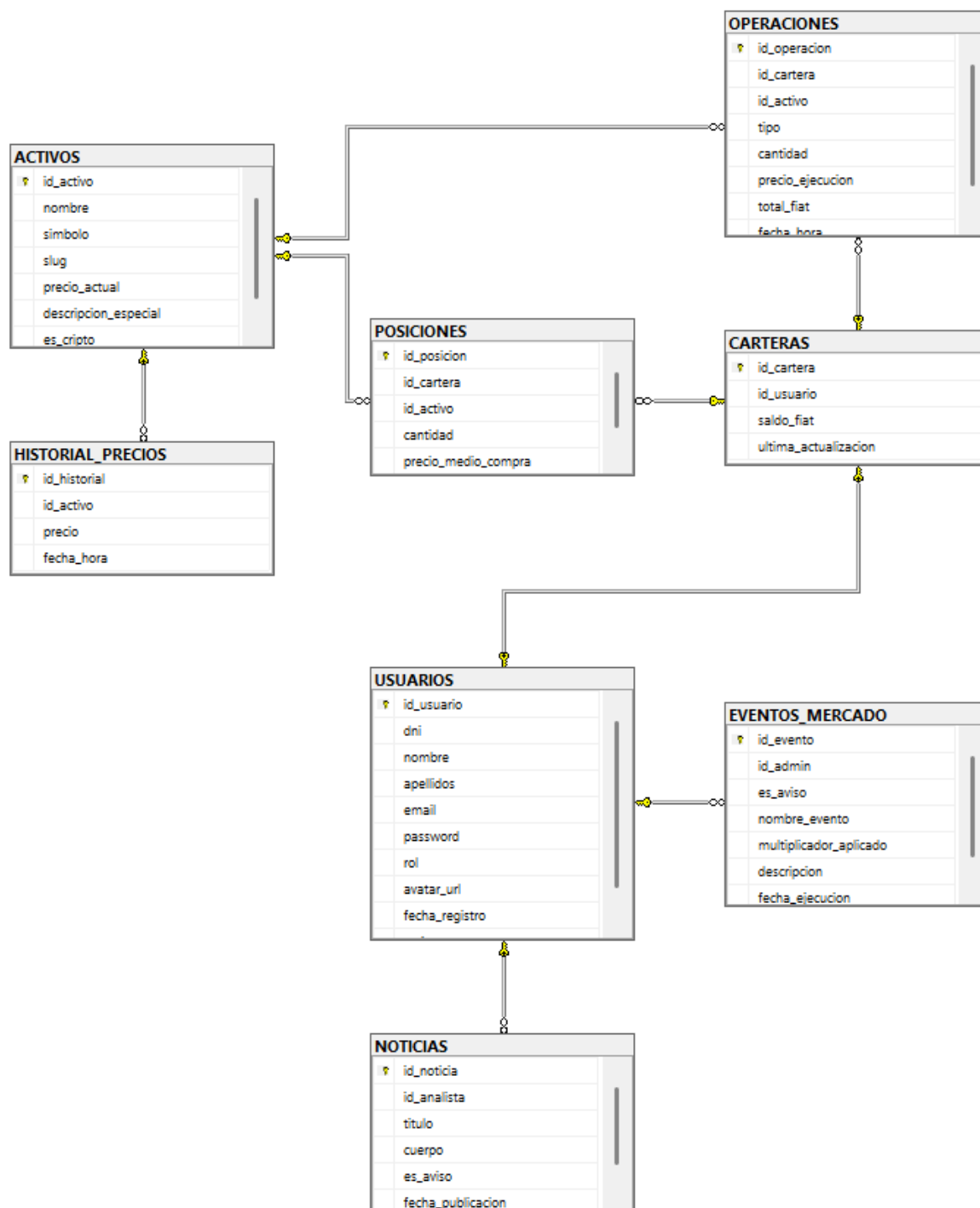


Figura 15: Esquema relacional